MUHAMMAD ZEIN AULIA

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SUMMARY

A student of Mechanical Design and Construction Engineering with a focus on engineering design and technical competencies in mechatronics, manufacturing, and control systems. Skilled in the design and analysis of machine components, creation and interpretation of technical drawings according to standards, and proficient in CAD software such as AutoCAD and SolidWorks. Knowledgeable in the integration of mechanical–electrical systems, fundamentals of actuators and sensors, and basic control system concepts. Possesses understanding of manufacturing processes, material selection, and design considerations for manufacturability. Accustomed to working analytically, meticulously, and with a focus on efficient, practical technical solutions.

PROJECT EXPERIENCE

Human-Powered Vehicle Design and Competition (Mechanical Team)

- Mechanical Assembly & Maintenance
- Drivetrain & Chassis Support
- Competition Technical Support

CVT Pulley Failure Analysis for Mio 2008

- Dimensional Measurement & Inspection
- 3D Modeling & Reconstruction
- CAE & Stress Analysis
- Failure Analysis

Reverse Engineering Powder Metallurgy Mixer

- Reverse Engineering & 3D Modeling
- Structural Analysis (CAE)
- CAE & Stress Analysis
- Electrical & Control System Analysis

PLC Material Handling System

- Electrical Wiring
- PLC Logic Development & Implementation
- System Testing & Troubleshooting

Mechanical Chopper Design and Prototype

- Manufacturing Process & Fabrication
- Component Purchasing
- Standard Part Selection
- Mechanical Assembly & Testing

Design and Prototyping Automatic Water Pump

- Planning & Concept
- Mechanical Design & 3D Modeling
- Manufacturing & Fabrication
- Mechanical Assembly
- Control System implementation

EDUCATION

Politeknik Negeri Bandung — **Agust 2023-Now**
Mechanical Engineering Study Program Mechanical Design and Contruction

MAN 1 Tasikmalaya — **July 2020-June 2023**
MIPA

Pesantren KHZ Musthafa Sukamanah — **July 2020-June 2023**
Ibtidaiyyah

HARDSKILL

- Mechanical Design & Drafting
- CAE / Simulation & Analysis
- Manufacturing & Fabrication
- Mechatronics & Electrical Systems
- Structural & Load Optimization
- Prototyping & Assembly

CERTIFICATE

- SOLIDWORKS CAD Design Associate
- SOLIDWORKS CAD Design Profesional
- Awareness Ahli K3 Umum
- Mini Class PLC Siemens
- Mini Class PLC Mitsubishi FX2N

SOFTWARE

- AutoCAD
- SolidWorks
- Inventor
- Mastercam X5
- Cimatron E
- MS Office

PORTOFOLIO